AI-POWERED SMART GYM SYSTEM

Sprint Plan & Product Backlog

Must-Have and Should-Have Backlog Items · Organised by Subsystem and Sprint

Derived from the Requirements Engineering specification, translating every Must and Should requirement into an ordered, dependency-aware two-week sprint cadence for a six-student, one-academic-year build.

11 sprints · 2 weeks each · 18 development weeks (Sprints 1 & 3 and 2 & 4 run in parallel)

Mobile application and computer vision: **Flutter**

Database: **Google Cloud Firestore**

Wearable: **ESP32 + MAX30102 + MPU6500 over Bluetooth Low Energy**

1. Introduction

1.1 Purpose

This document sequences the Must- and Should-priority requirements of the AI-Powered Smart Gym System's Requirements Engineering specification into a delivery plan: eleven two-week sprints for a six-student team working across one academic year. Its purpose is to state, for each sprint, what will be built and why that ordering was chosen — so that a Must item is never scheduled after something that depends on it, and so that supervision, planning and progress tracking share one agreed sequence.

The team's two subteams run Sprints 1 and 3 in parallel during weeks 1–2, and Sprints 2 and 4 in parallel during weeks 3–4 — one subteam on the Auth/Onboarding-then-Data-Layer track, the other on the Computer Vision track — since neither pair depends on the other's output. From week 5 onward every remaining sprint depends on both tracks being in place, so the whole team works through Sprints 5–11 sequentially. This compresses the schedule from eleven sequential slots to nine two-week windows, or 18 development weeks in total.

Every backlog item carries the identifier of the requirement it realises, exactly as numbered in the Requirements Engineering document (Sections 3 and 4). This sprint plan carries no independent numbering scheme for scope: an item's ID is a direct cross-reference, and if a requirement is renumbered, added or removed in that document, this plan is updated to match rather than kept as a separate source of truth.

This revision has been synced to the current Requirements Engineering document. Items dropped from that document (superseded or renumbered) have been removed from the backlog below; FR-NUT-08 has been removed as it is now Could-priority, which is out of scope for this plan; and NFR-SEC-02 (automated Security Rules test coverage) has been added to Sprint 2, since it now appears as a Must-priority requirement with no prior backlog item. Section 3 adds a new Flutter client-side implementation plan, mapping each sprint's backlog to concrete app architecture, packages and modules.

1.2 Scope

In scope for this document: the assignment of every Must- and Should-priority functional and non-functional requirement to one of eleven sprints, an overview table giving each sprint's duration and goal, a table of backlog items ordered by priority within each sprint, and a Flutter client-side implementation plan (Section 3).

Out of scope: Could-priority requirements, detailed task breakdown below the backlog-item level, story-point or effort estimation, and individual task assignment within the team's two subteams — these are maintained separately and are not repeated here.

2. Sprint Plan

2.1 Overview

Eleven two-week sprints across nine two-week windows. Sprints 1 & 3 and Sprints 2 & 4 run in parallel across the team's two subteams; every later sprint runs with the whole team once both tracks are in place. Within this schedule, each subsystem's Must items land before its Should items, and before any later subsystem that depends on it.

Sprint	Weeks	Team	Goal
1	Weeks 1–2	Team A	Auth & Onboarding Foundation — Stand up accounts, sign-in and the onboarding flow that every later feature reads from.
3	Weeks 1–2	Team B	Computer Vision Core I — Capture & Pose — Get the camera stream, BlazePose and rep/angle math running on-device.
2	Weeks 3–4	Team A	Data Layer & Sync — Build the Firestore schema, security rules and offline sync every subsystem will write to.
4	Weeks 3–4	Team B	Computer Vision Core II — Feedback & Summaries — Turn pose data into live coaching cues, the skeleton overlay and per-set summaries.
5	Weeks 5–6	Whole team	Session Management & Safety — Wrap the CV engine in a structured workout session with rest timers and safety alerts.
6	Weeks 7–8	Whole team	Wristband & Biometric Integration — Bring the ESP32 wristband online over BLE and pipe biometrics into the app.
7	Weeks 9–10	Whole team	Egyptian Nutrition & Meal Planning — Ship the food catalogue, search, meal-plan generator and logging.
8	Weeks 11–12	Whole team	Bilingual Conversational Assistant — Add the domain-restricted chatbot with provider failover and safe context handling.
9	Weeks 13–14	Whole team	Localisation & Accessibility — Complete the bilingual UI, RTL mirroring and accessibility baseline.
10	Weeks 15–16	Whole team	Analytics, Scalability & Reliability Hardening — Add progress charts and validate the system against its scale and reliability targets.
11	Weeks 17–18	Whole team	Compliance, Security Finalisation & Release Readiness — Close out legal, security and packaging work and confirm the whole system fails gracefully.

2.2 Sprint 1 — Auth & Onboarding Foundation

Weeks 1–2 · Team A. Goal: Stand up accounts, sign-in and the onboarding flow that every later feature reads from.

ID	Backlog Item	Priority
FR-AUTH-01	Build the email/password sign-up flow that auto-creates the Firestore profile document at /users/{uid} with the required fields.	Must
FR-AUTH-02	Add email/password and Google Sign-In, resolving both to the same Firebase UID when the verified email matches.	Must
FR-AUTH-03	Implement a self-service password-reset flow that never reveals whether an email address is registered.	Must
FR-AUTH-04	Enforce the password policy (8–32 chars, printable ASCII + space) and let Firebase Auth handle hashing/storage.	Must
FR-AUTH-05	Persist the authenticated session across restarts.	Must
FR-PROF-01	Build an onboarding flow (≤ 8 screens) that collects the eight profile attributes.	Must
FR-PROF-02	Compute and store BMI, refreshed on every height/weight update.	Must
FR-PROF-03	Implement Resting Energy Expenditure via the Mifflin-St Jeor equation.	Must
FR-PROF-04	Derive TDEE and protein/carb/fat targets from REE, activity factor and goal.	Must
FR-PROF-07	Add an in-app Arabic/English language switch that applies without a restart.	Must
NFR-SEC-01	Enforce TLS ≥ 1.2 and certificate pinning on backend endpoints.	Must
NFR-SEC-03	Enforce the NIST-aligned password policy, including a compromised-password check.	Must
NFR-MNT-01	Set up the layered architecture (presentation/domain/data) with no Firestore/camera/plugin dependency in the domain layer.	Must
NFR-COMP-01	Ship one Flutter codebase with documented Android/iOS parity exceptions only.	Must
NFR-COMP-02	Support Android 8.0+ (API 26) and iOS 13+.	Must
FR-AUTH-06	On sign-out, clear the local Isar cache of personal data, biometrics and chat history, keeping only reference data.	Should
FR-AUTH-07	Add an account-deletion flow that recursively removes the /users/{uid} document tree and the Auth record.	Should
FR-PROF-05	Estimate HRmax and derive the five training-intensity zones.	Should
FR-PROF-06	Set a default weekly activity target from activity level and goal, user-overridable.	Should

2.3 Sprint 3 — Computer Vision Core I — Capture & Pose

Weeks 1–2 · Team B. Goal: Get the camera stream, BlazePose and rep/angle math running on-device.

ID	Backlog Item	Priority
FR-CV-01	Wire up a live camera stream at ≥ 720p/30fps.	Must
FR-CV-02	Integrate on-device BlazePose to extract 33 landmarks and world landmarks per frame.	Must
FR-CV-03	Support Lite/Full/Heavy model variants with device-tier auto-selection and a manual override.	Must
FR-CV-04	Compute joint angles (knee, hip, trunk, elbow, shoulder) in Dart via 2-D vector trigonometry.	Must
FR-CV-05	Implement per-exercise repetition counting using hysteresis thresholds.	Must
FR-CV-06	Build the posture risk-scoring rule set (knee valgus, ROM/depth, lumbar flexion, forward lean, elbow flare, asymmetry).	Must
NFR-PERF-01	Keep on-device pose inference ≤ 30 ms/frame, sustaining ≥ 25 fps during an active set.	Must
NFR-PERF-10	Keep camera preview latency (movement to screen) ≤ 100 ms.	Must
NFR-COMP-03	Require ≥ 3 GB RAM + 720p/30fps camera for pose coaching; disable cleanly below spec.	Must
NFR-SEC-04	Guarantee no raw camera frame, image or video ever leaves the device, including crash/diagnostic uploads.	Must

2.4 Sprint 2 — Data Layer & Sync

Weeks 3–4 · Team A. Goal: Build the Firestore schema, security rules and offline sync every subsystem will write to.

ID	Backlog Item	Priority
FR-DATA-01	Stand up the Firestore collection structure (/users, /workouts, /sets, /biometrics, /meals, /chat, /foods, /exercises, /appConfig, /auditLog).	Must
FR-DATA-02	Give every document a schemaVersion integer and a serverTimestamp updatedAt field.	Must
FR-DATA-03	Enable Firestore offline persistence — cached reads, durably queued offline writes.	Must
FR-DATA-04	Aggregate biometric samples into fixed time-window documents rather than one per sample.	Must
FR-DATA-05	Use batched writes/transactions for multi-document updates, within the per-commit limit.	Must
FR-DATA-06	Write deny-by-default Security Rules scoping each client to its own /users/{uid} subtree.	Must
NFR-SCAL-02	Keep reads/writes at 500 DAU (Daily Active Users) inside the Firestore free tier.	Must
NFR-SCAL-04	Model unbounded lists as subcollections to avoid the 1 MiB document limit.	Must

ID	Backlog Item	Priority
NFR-SEC-02	Cover the deny-by-default Security Rules with an automated test suite, including cross-user.	Must
NFR-PERF-12	Commit biometrics as one aggregated document per 60 s window per user, never per sample.	Must
FR-DATA-07	Mirror food/exercise catalogues into local storage at launch for offline search, refreshed on version mismatch.	Should
FR-DATA-08	Define composite indexes in advance for every planned query.	Should
NFR-SCAL-01	Confirm the architecture supports 5,000 accounts / 500 DAU without change.	Should
NFR-PERF-05	Acknowledge writes locally at once; commit to Firestore within 500 ms p95 on 4G.	Should

2.5 Sprint 4 — Computer Vision Core II — Feedback & Summaries

Weeks 3–4 · Team B. Goal: Turn pose data into live coaching cues, the skeleton overlay and per-set summaries.

ID	Backlog Item	Priority
FR-CV-07	Deliver real-time corrective feedback via synchronised TTS and on-screen overlay, rate-limited to one cue per 2s.	Must
FR-CV-08	Render the colour-coded (green/amber/red) skeleton overlay on the camera preview.	Must
FR-CV-09	Discard low-visibility landmarks; suspend scoring and prompt repositioning after 1s of missing landmarks.	Must
FR-CV-10	Add a pre-set framing calibration with a 3-second audible countdown before scoring begins.	Must
FR-CV-11	Guarantee no raw frame, image, video or per-frame landmark stream is ever transmitted or persisted.	Must
FR-CV-12	Ship the v1 exercise library: squat, deadlift, bench press.	Must
FR-CV-14	Produce the per-set summary — reps, time-under-tension, ROM, tempo, form score, fault counts.	Must
NFR-PERF-02	Keep feedback latency ≤ 100 ms to the overlay and ≤ 300 ms to the spoken cue.	Must
NFR-PERF-08	Keep app start ≤ 3 s cold, ≤ 1.5 s warm, ≤ 1 s hot.	Must
FR-CV-13	Monitor inference frame rate and auto-downgrade model/resolution if it drops below the performance floor for 3s.	Should
NFR-PWR-05	Downgrade model/rate before the platform throttles, avoiding sustained thermal throttling in a 45-minute session.	Should

2.6 Sprint 5 — Session Management & Safety

Weeks 5–6 · Whole team. Goal: Wrap the CV engine in a structured workout session with rest timers and safety alerts.

ID	Backlog Item	Priority
FR-ALRT-01	Model a workout as an ordered exercise list with sets/reps/load and per-set completion tracking.	Must
FR-ALRT-02	Add an automatic inter-set rest timer.	Must
NFR-REL-02	Keep coaching, set logging, rest timer and food logging fully usable offline with zero data loss.	Must
FR-ALRT-03	Raise a caution when heart rate exceeds 90% of estimated HRmax.	Should
FR-ALRT-04	Show a warm-up recommendation and a mandatory first-session safety acknowledgement.	Should

2.7 Sprint 6 — Wristband & Biometric Integration

Weeks 7–8 · Whole team. Goal: Bring the ESP32 wristband online over BLE and pipe biometrics into the app.

ID	Backlog Item	Priority
NFR-COMP-04	Require BLE 4.2+ for the wristband link; its absence disables only biometric features.	Must
NFR-REL-04	Re-establish the wristband link within 10 s of returning to range.	Should
NFR-PWR-01	Target ≥ 8 h continuous session use per charge.	Should
NFR-PWR-06	Have the wristband enter deep sleep when battery voltage stays ≤ 2.95 V for 5 consecutive readings.	Should
NFR-PERF-04	Surface a biometric sample in the UI within 200 ms of its generation on the band.	Should
NFR-PERF-11	Sample the optical sensor at ≈ 100 sps while the band is connected.	Should

2.8 Sprint 7 — Egyptian Nutrition & Meal Planning

Weeks 9–10 · Whole team. Goal: Ship the food catalogue, search, meal-plan generator and logging.

ID	Backlog Item	Priority
FR-NUT-01	Load the curated Egyptian food catalogue (≥ 300 entries, named staples).	Must
FR-NUT-02	Give every food entry bilingual names, category, per-100g macros and ≥ 1 household portion.	Must
FR-NUT-04	Implement fuzzy Arabic/English/transliterated food search.	Must
FR-NUT-05	Build the daily meal-plan generator (3 meals + ≤ 2 snacks, TDEE ± 5%, protein ± 10%).	Must
FR-NUT-07	Add food logging (search / portion / repeat-yesterday) with running daily totals.	Must
FR-NUT-09	Record dietary exclusions and enforce them in every generated plan.	Must

ID	Backlog Item	Priority
NFR-DQ-04	Ensure ≥ 95% of catalogue entries have complete macros and 100% carry a source reference.	Must
FR-NUT-03	Attach an indicative EGP price with a priceAsOf date; flag entries stale after 90 days.	Should
FR-NUT-06	Show estimated plan cost in EGP; allow regeneration under a stated daily budget ceiling.	Should
NFR-DQ-05	Prefer the national composition table over a generic international one for indigenous dishes.	Should
NFR-PERF-09	Return local food search results within 500 ms of the last keystroke.	Should

2.9 Sprint 8 — Bilingual Conversational Assistant

Weeks 11–12 · Whole team. Goal: Add the domain-restricted chatbot with provider failover and safe context handling.

ID	Backlog Item	Priority
FR-CHAT-01	Detect the message language and reply in matching MSA / Egyptian Arabic / English.	Must
FR-CHAT-03	Constrain the assistant to fitness/nutrition/safety topics; decline out-of-domain requests politely.	Must
FR-CHAT-04	Inject goal, experience, limitations, recent workout summary and macro targets as context (never raw biometric series).	Must
FR-CHAT-06	Refuse medical diagnosis/prescription and show a persistent non-diagnostic disclaimer.	Must
FR-CHAT-07	Persist chat history (role, content, language, model, tokens, timestamp) under /users/{uid}/chat, with a clear-history option.	Must
NFR-SEC-05	Proxy all LLM calls through the backend; embed no third-party API credentials client-side.	Must
NFR-SEC-06	Strip direct identifiers from chat text before transmission, disclosed in the consent flow.	Must
NFR-SEC-11	Rate-limit chat completions server-side, per user and per IP.	Must
NFR-REL-05	Fail over to the secondary LLM provider if no first token arrives within 5 s.	Must
FR-CHAT-02	Add primary-to-secondary LLM provider failover on error, timeout or rate-limit.	Should
FR-CHAT-05	Stream the reply to the client.	Should
FR-CHAT-08	Return a cached/templated fallback for common intents when both providers are unavailable.	Should
NFR-SEC-09	Write a documented breach-response procedure.	Should
NFR-PERF-07	Start streaming a chat reply within 1.5 s p95; finish a 150-token reply within 3 s.	Should

2.10 Sprint 9 — Localisation & Accessibility

Weeks 13–14 · Whole team. Goal: Complete the bilingual UI, RTL mirroring and accessibility baseline.

ID	Backlog Item	Priority
FR-LOC-01	Build the complete Arabic + English UI.	Must
FR-LOC-02	Fully mirror layout — nav, icons, charts, lists — in RTL for Arabic.	Must
FR-LOC-03	Localise numerals and format dates/time/currency to the Egyptian locale.	Must
FR-LOC-05	Replace raw exceptions/stack traces with localised, actionable error messages.	Must
NFR-USE-04	Meet ≥ 4.5:1 text contrast in both themes and both languages.	Must
NFR-USE-05	Keep every pointer target ≥ 44×44 logical pixels.	Must
FR-LOC-04	Record spoken coaching cues in both languages, Arabic in Egyptian phrasing.	Should
NFR-USE-06	Have ≥ 4 native reviewers judge the Arabic UI as natural, not a literal translation.	Should
NFR-USE-07	Verify coaching cues are intelligible over gym noise, with a visual counterpart for every audio cue.	Should
NFR-USE-08	Keep every screen's primary task readable in ≤ 40 words.	Should

2.11 Sprint 10 — Analytics, Scalability & Reliability Hardening

Weeks 15–16 · Whole team. Goal: Add progress charts and validate the system against its scale and reliability targets.

ID	Backlog Item	Priority
FR-ANL-03	Chart mean form score per exercise over time.	Should
FR-ANL-04	Chart HR trends per session and resting HR over time, from stored aggregates.	Should
FR-ANL-05	Track adherence — sessions completed vs. planned, minutes vs. weekly target.	Should
NFR-DQ-01	Validate reported HR against a chest-strap reference within ± 10 bpm for ≥ 90% of paired samples.	Should
NFR-DQ-03	Validate rep-counting (≥ 95%) and fault-classification (≥ 90%) accuracy against labelled ground truth.	Should
NFR-SCAL-07	Confirm search stays fast at 2,000 food / 100 exercise catalogue entries.	Should
NFR-SCAL-08	Cap and prune chat history per user to bound storage growth.	Should
NFR-REL-01	Track the cloud data tier against the ≥ 99.9% monthly availability target.	Should
NFR-USE-03	Keep starting a tracked set from the home screen to ≤ 3 taps.	Should

2.12 Sprint 11 — Compliance, Security Finalisation & Release Readiness

Weeks 17–18 · Whole team. Goal: Close out legal, security and packaging work and confirm the whole system fails gracefully.

ID	Backlog Item	Priority
NFR-LEG-01	State prominently, at onboarding and in settings, that this is a wellness aid, not a medical device.	Must
NFR-LEG-02	Base data processing on explicit, per-purpose consent, with access/correct/erase/object/withdraw support.	Must
NFR-LEG-03	Handle health data as a special category with enhanced safeguards.	Must
NFR-LEG-05	Keep coaching/nutrition guidance within general-wellness advice; decline injury/condition-specific advice.	Must
NFR-REL-09	Verify no single failure (band, network, either LLM provider, camera permission) cascades beyond its own feature.	Must
NFR-SEC-07	Retain biometric window documents ≤ 12 months, then aggregates only.	Should
NFR-SEC-08	Document the database region and disclose any cross-border transfer.	Should
NFR-SEC-12	Log all admin actions on shared catalogues to an append-only, user-unwritable audit log.	Should
NFR-LEG-04	Ship an open-source notices screen honouring third-party licence obligations.	Should
NFR-LEG-06	Keep activity recommendations within WHO weekly-minute guidance.	Should
NFR-MNT-02	Reach ≥ 70% domain-layer unit-test coverage.	Should
NFR-MNT-03	Keep exercise rules/cues as declarative configuration, not branching code.	Should
NFR-MNT-05	Document a one-command reproducible build per component.	Should
NFR-MNT-06	Make diagnostic logging structured, off by default in release, and free of personal/biometric data.	Should
NFR-PERF-13	Keep the Android bundle ≤ 60 MB including the bundled pose model.	Should